

PS 5052 Mathematical Modeling in Political Science

Fall 2016

This course is an introduction to mathematical techniques used to model phenomena studied in political science, with special attention to the analysis of individual action. Mathematical topics covered include: sets, functions, and graphs; matrix algebra; differential calculus and optimization; probability and risk; integral calculus; and sequences, series, and limits. All these topics are useful in many settings in political science, including game theory, dynamic modeling, and statistics.

This course website will be updated to reflect any changes in schedule, topics covered, or assignments, as well as to provide relevant links to materials associated with the course.

Course Outline and Approximate Schedule

o. Optional pre-semester basic math review sessions

Tue 8/23 and Thu 8/25 2:00-3:00 pm in Siegle 205

Mainly, these sessions will be devoted to your questions on the basic math review problems, previously distributed. **You should try to do the problems in at least the first three Sections before you come to the first session.**

The Assessment problems are for my information, and those topics will be covered during the semester. **All students are to hand in their answers (or "don't knows") to the Assessment problems on the first day of class, Monday 8/29.**

[Click here](#) for another complete copy of Basic Review and Assessment problems. Solutions to the basic review are posted below, section-by-section.

- Solutions for [Section 1, Algebra](#)
- Solutions for [Section 2, Sets](#)
- Solutions for [Section 3, Functions and Graphs](#)
- Solutions for [Section 4, Statements and Proofs](#)

Also for the review sessions: you should try to read the following sections in the Pemberton and Rau text. I'll take any questions you may have on those chapters.

- Chapter 1, sections 1.1, 1.3,
- Chapter 2, sections 2.1
- Chapter 3, sections 3.1-3.4
- Chapter 4, sections 4.1-4.4
- section 31.1 "Rigour" (pp. 700-702)

Further relevant material you may find helpful in the supplementary texts

- on algebra: Gill sections 1.2-1.4 and 1.7; Moore & Siegel chapter 2
- on sets: Gill section 7.3; Moore & Siegel sections 1.1-1.2
- on functions and graphs: Gill section 1.5 and 2.1-2.2; Moore & Siegel section 3.1
- on statements and proofs: Moore & Siegel section 1.7

Additional exercises or examples on equations involving exponentials and logarithms:

- There are several such exercises at the end of Section 17 in Gill's textbook.
- [This page](#) and especially [its following pages](#) from the website "Purplemath" offers several examples and solution techniques.
- Try the examples and problems in [Tutorial 45](#) and [Tutorial 46](#) on the "Virtual Math Lab" site of W. Texas A&M University.

1. Sets and relations, with applications to choice theory

M 8/29 to W 8/31: Relations on sets; preference and choice

Text: [[Lecture notes on Relations, Preference, and Choice](#)]

- sets; sets of real numbers
- relations on sets
- application: choice theory

- partial orders
- **Exercises due Wednesday 9/7** all those from the Lecture Notes on Relations, Preferences, and Choice.

Special Friday software session 9/2: Mathematical Typesetting with LaTeX

- [How to Install LaTeX](#): Please try this prior to the session
- all subsequent homework should be typeset in LaTeX

M 9/12 Special software session: Programming with R

2. Sequences and series; limits and continuity

W 8/31 to W 9/21: Analysis

Text: P&R, selections (below) from chapters 5, 6, 31; for outside reading, Moore & Siegel ch. 3

You should read sections 5.1 by Wednesday 8/31, the remainder of Chapter 5 by Wednesday 9/7

- real numbers, sequences, limits (Pemberton and Rao 5.1, 31.2, 31.3)
- sums and series (P&R 5.2; see also Gill, pp. 259-266)
- **application**: choices over time (P&R 5.3)
- continuous functions (P&R 5.4; Section 31.4)
- Fixed-point theorems (recommended advanced reading: P&R section 32.4)
- **Exercises due Monday 9/12—Please hand in to Dave**
 - 5.1.3
 - 5.2.1
- **Exercises due Monday 9/19**
 - 5.2.2, 5.2.3
 - 5.3.2, 5.3.4
 - 5.4.1, 5.4.2
 - and this exercise using R:

- Use the "function" command to create a list consisting of the first 1000 elements of the sequence (u_n) defined by $u_n = 1 + 1/n$. Use the clipboard to copy your method and the sequence values from the console to a page in an editing program such as Microsoft Word.
- Plot the values of the sequence (vertical axis) against the values of n from 1 to 1000 using the "plot" or other graphing command.

3. Differential calculus

Text: P&R chapters 6-8.

You should read sections 6.1-6.3 for Monday, 9/19

Homework problems due Monday 9/26:

- All problems listed below from Chapter 6
- An R-programming exercise:
 - In a single graph, plot the function $y = f(x) = x^3 - 2x^2$ (where "^" indicates an exponent) and its tangent line at $x = 1.62$. (Let the graph cover x values from -3 to 3.)
 - In a single graph, plot the same function $y = f(x)$ and its derivative $y = f'(x)$.

M 9/19, W 9/21, and M 9/26: The derivative

- Read by Monday 9/19: P&R sections 6.1-6.3.
- Read by Wednesday 9/21: Sections 7.1, 7.2
 - Exercises
 - 6.1.1, 6.1.3
 - 6.2.1, 6.2.3
 - 6.3.1 through 6.3.3
 - The latter problem refers to "the answer to Exercise 5.4.4." Do that also, as needed to answer 6.3.3. Exercise 5.4.4 requires you to apply the intermediate value theorem.
- Methods of differentiation
 - **Exercises due Monday 9/26**
 - 7.1.1 and 7.1.2
 - 7.2.1 through 7.2.3
- Monotone functions and inverse functions: read by Monday 9/26 Sections 7.3; 7.4 up to the "Application: inverse demand functions"

- **Exercises:** bring, with your questions, to review session on Wed. 9/28; not to turn in
 - 7.3.1, 7.3.2
 - 7.4.1

First Exam 9/28 - 10/3

- Wed. Sept. 28: Review
- Wed. or Fri.: distribute exam
- Exam due at beginning of class, Mon. 10/3

M 10/3 and W 10/5 Using the derivative

Homework problems due Monday 10/10:

- 7.3.1, 7.3.2
- 7.4.1
- 8.1.1 through 8.1.3
- 8.2.1 through 8.2.3
- 8.3.1, 8.3.2, 8.3.4

Monotone functions and inverse functions:

- P&R section 7.3-7.4. Exercises:
- 7.3.1 and 7.3.2
- 7.4.1

Finding Maxima and minima:

- P&R sections 8.1-8.3. Exercises:
- 8.1.1 through 8.1.3
- 8.2.1 through 8.2.3
- 8.3.1, 8.3.2, 8.3.4

M 10/10 Convexity and concavity

- P&R section 8.4. Exercises:
 - 8.4.1 through 8.4.6

Below this point, calendar dates have not been updated from 2015 -- ignore all

4. Applying differentiation to exponentiation, logarithms, approximation, and limits

Text: P&R chapters 9-10

HW problems due Monday 10/19

- 8.4.1 through 8.4.6
- 9.1.4, 9.2.2, 9.2.4
- 10.1.1, 10.1.2
- **R-exercise for Newton's Method:** Use Newton's method to approximate roots of the function $y = x^2$. In a single graph, plot the function and its tangent line at each guess of the root. (Let the graph cover x values from -1 to 1.) Begin by guessing a root at 1. Proceed through Newton's method, generating new approximations to the root. Go through two updates from your initial guess, and write out the results to the console or to a file.

Monday 10/12: The exponential function

P&R sections 9.1, 9.2, and Appendix.

Homework exercises: 9.1.4, 9.2.2, 9.2.4

Illustrations

- Spreadsheet illustrating the approximation of $\exp(x)$

Wednesday 10/14: Newton's method of approximation; L'Hôpital's rule

P&R sections 10.1 and pp. 170-172 of 10.2.

Homework exercises: 10.1.1, 10.1.2; for next week, 10.2.3

Examples and Illustrations

- To be posted: a spreadsheet illustrating the example calculation using Newton's method in Pemberton & Rao, p. 168-169. You can plug in different starting values, which I label " x_0 ".
- There are a number of nice graphic illustrations of Newton's method online, such as [as this one by Paul Garrett](#). Follow the link to his illustration of an important pathological case. (If you're interested, I can supply a spreadsheet illustrating Garrett's example. You can fill in the constant term in the polynomial: $K = 0.3$ (default) to get the well behaved case, or $K = 0.4$ to get the pathological case.)
- [Here is another](#) nice example from Wikipedia, as well as [another in which the Newton method may fail](#) to converge. In the latter, the derivative at $x = 0$ is -1 , the derivative at $x = 1$ is 1 , and both tangent lines take you back to the other point, so that $x_n = 0, 1, 0, 1, \dots$, although the only true root is somewhere else entirely.

5. Linear algebra

Text: P&R chapters 11-13, plus (if time) some material from 25, 26.

HW problems due Monday 10/26

- 10.2.3 on L'Hopital's rule
- all exercises listed below from Chapter 11

Mon. 10/19 & Wed. 10/21: Vectors and Matrices

- P&R, chapter 11
- Turn in the following exercises
 - 11.1.4 (a-f) (p. 211)
 - 11.2.3-11.2.5 (pp. 215-16)
 - 11.3.1 and 11.3.2 (pp. 219-220). In 11.3.2, what's called y in the problem is labeled w back in equation (11.3); they're supposed to be the same thing.
 - 11.4.2, 11.4.4 (p. 222)
 - and more below from chapter 12

HW problems due Monday 10/26

- all exercises listed from Chapter 12
- all exercises from sections of Chapter 13 covered in class

Mon. 10/26: Systems of Linear Equations

- P&R, chapter 12
- **Exercises to turn in Monday, 11/2:**
 - 12.1.2 (a-e) (p. 229)
 - 12.2.1 (a-c) (p. 234)
 - plus more below

Wed. 10/28: Quadratic Forms

- P&R, chapter 13
- **More exercises to turn in Monday, 11/2:**
 - 12.3.1, 12.3.2, 12.3.3, 12.3.4 (p. 240)
 - 13.1.1 (p. 252)
 - exercise 13.4.2, 13.4.7 (p. 264-65)
 - Multiply out the quadratic form $\mathbf{x}^T \mathbf{A} \mathbf{x}$ in variables x_1, x_2 and x_3 defined by the following 3x3 matrix:

$$\mathbf{A} = \begin{bmatrix} 4 & 2 & 1 \\ 2 & 3 & 0 \\ 1 & 0 & 5 \end{bmatrix}$$

(For a helpful alternative treatment of topics in matrix analysis, see the following recommended supplementary readings on vectors and matrices from Gill:

- Chapter 3: 3.1-3.5 (omit examples 3.21, 3.22)
- Chapter 4: 4.3, 4.4, 4.6, 4.7, 4.9 (except example)).

6. Multivariate calculus

Text: P&R chapters 14-17

Exercises to turn in Monday, 11/9:

- 14.1.1 (a-d) (p. 275)
- 14.2.2 & 14.2.3 (p. 281)
- Exercises "15.A" and "15.B," below
- 16.1.5 (three parts)

Mon. 11/2: Multivariate functions and partial derivatives

- P&R, chapter 14.1, 14.2
- Exercises:
 - 14.1.1 (a-d) (p. 275)
 - 14.2.2 & 14.2.3 (p. 281)

Wed. 11/4 Implicit functions; optimization

- P&R, chapter 15.1; 16.1, 16.2
- Exercises :
 - For material in Chapter 15, do the following exercises:
 - "15.A": Define $F(x,y) = x^2 + y^2$, and suppose that $F(x,y) = t^2$, where $t > 0$ is a parameter.
 - a. Express y as an explicit function of x and t .
 - b. Using your formula, find a point (x_0, y_0) satisfying $F(x,y) = t^2$, and evaluate dy/dx at that point.
 - c. Use the rule of implicit differentiation to find the same result. Show your work.
 - d. In the neighborhood of (x_0, y_0) , find how y varies as t varies: that is, evaluate the derivative of y with respect to t .
 - "15.B": Suppose a decision maker faces this problem: a given level x of product safety regulation provides social benefits $b(x)$ and imposes economic costs $c(x; t)$, where $t > 0$ is a parameter measuring competition in the relevant industry; costs are higher for higher t . The decision maker chooses x (nonnegative) to maximize $b(x) - c(x; t)$.
 - a. Suppose in particular that $b(x) = 4x$ and $c(x; t) = tx^2$. Solve the maximization problem and use derivatives to show how the decision maker's choice x will change when t increases.
 - b. Rather than use the specific functional forms in (a.), simply assume that benefits increase in x at a decreasing rate; and that costs increase in x at an increasing rate. Give these assumptions in the form of conditions on partial derivatives, and use implicit differentiation to answer the same comparative statics question as in part (a.).
 - 16.1.5 (p. 326)
 - for later: 16.2.2, 16.2.6 (p. 334-35)†

Mon. 11/9 Optimization with constraints: the method of Lagrange

- P&R, chapter 17.1-17.2
- **Exercises to bring to class on Wednesday 11/11 for review session: 17.1.1**
 - Hint: In order to "show how you know the value you have found is a maximum," draw a graph of the level curves of xy , and show the direction in which xy increases; plot the constraint on the same graph; and explain the answer using your graph.

Second Exam

To cover everything from monotone functions, inverse functions, optimization with one variable, convexity and concavity, the exponential function, Newton's method of approximating roots, and L'Hopital's rule, as well as all our material on matrix analysis and everything covered up until the exam on the calculus of functions of several variables.

Wed. 11/11 Review

exam handed out Wed. 11/11, due at beginning of class, Mon. 11/16.

7. Probability and Mathematical Statistics

Text: "Probability," to be distributed, along with the corresponding LaTeX script.

MW 11/16-18: Probability Theory and Expected Utility

Exercises to turn in Monday, 11/23:

- Exercises on probability theory, to be distributed; all problems except the last one in Section D.

8. Integral calculus

Text: P&R sections 19.1, 19.2; chapter 20 as time permits.

M 11/23, M 11/30, W 12/2: Integral Calculus

Material from Pemberton & Rao:

- 19.1 area under a graph; the anti-derivative; the Fundamental Theorem of Calculus
- 19.2 indefinite and definite integrals
- 20.1 methods of integration (briefly)
- 20.2 "infinite" integrals; "improper" integrals
- briefly, if time: 20.4, double integrals

Exercises to be brought to class for discussion in final review session:

- The final exercise from the "exercises in probability theory" handout
- 19.1.3 and 19.1.4 (p. 404)
- 19.2.2 and 19.2.5 (p. 409-10)
- 20.1.1 (a & b, p. 429) and 20.2.1 (a-c, p. 432)
- The following problem, "number 8": Suppose X is a random variable whose density function is $f(x) = x^2$ for x in the interval $[0, a]$, and $f(x) = 0$ for x outside that interval. Show your work in answering the following questions:
 - a. Using the fact that the whole area under the density function must be equal to 1, find the exact value of a . (Hint: the answer is approximately $a = 1.44$.)
 - b. Derive a formula that gives the value, $F(x)$, of the CDF for every real number x . (Hint: you're on your own; does the answer to part c come out right?)
 - c. Use your answer to part b to find $\Pr(.5 < X < 1)$. (Hint: the answer is approximately 0.292.)
 - d. Integrate to find EX . (Hint: the answer is approximately 1.075)

Third Exam

Covering probability theory and integral calculus.

Mon. 12/7 review session in Seigle 106

Exam to be distributed Tuesday morning 12/8

Answers due: Sunday 12/13 10:00 am

This page written by Randall Calvert 2015
Email comments and questions to calvert at wustl.edu

PS 5052

- [Home](#)
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Monday & Wednesday 10:00-11:30
classroom: Seigle 204

[Jump directly to current topics & assignments](#)

Teaching Assistant

Dave Carlson

WU email address carlson.david

Office Hours: TBA

Optional Help Sessions: Fridays 10:00-11:30 in **Seigle 111**

Textbooks

Course Requirements

Arts & Sciences | Washington University in St. Louis | One Brookings Drive, St. Louis, MO 63130-4899